

Migration mechanisms of oil-water-sand three-phase flow in unconsolidated sandstones during water flooding: An image analysis approach resilient to threshold interference

Maozhu Li, Yiqiang Li, Zhipeng Wang, Xinji Du, Lei Wang, Jinxin Cao , Zheyu Liu ^{*}

National Key Laboratory of Petroleum Resources and Engineering, China University of Petroleum (Beijing), 102249, PR China

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ABSTRACT

Unconsolidated sandstone reservoirs exhibit high sand production susceptibility during water flooding, impairing fluid migration and well productivity. Elucidating three-phase (oil-water-sand) migration mechanisms and residual fluid distribution characteristics is critical for sanding rate prediction and fluid saturation estimation. This study examines sand migration and residual oil distribution during waterflooding in unconsolidated sandstone oil reservoirs. Using methylene blue-dyed water to flood artificial models, we replicated sand production patterns and quantified outcomes via novel image analysis. Results show that: In water flooding processes through heterogeneous unconsolidated sandstones, sand production rate exhibits a quadratic polynomial relationship with time. Greater fluctuation ranges in areal sweep variation rates indicate frequent flooding front diversions. Post-flood residual oil saturation measures 66.07–71.87 % in heterogeneous models versus 70.99–76.65 % in homogeneous counterparts. Residual oil saturation demonstrates broader distribution in swept zones of heterogeneous formations, whereas homogeneous systems show localized concentration. Strategic injection placement in high-permeability zones induces enhanced sweep efficiency when fluid propagates into low-permeability regions, establishing optimized drainage networks. These insights aid sand control and residual fluid management during fluid injection/production. The image analysis method enables pore-scale quantitative characterization for aquifer management and contaminant remediation.

1. Introduction

Unconsolidated sandstone formations are widely distributed globally and often serve as crucial aquifers, geothermal reservoirs, or pathways for contaminant migration (Chen et al., 2023, Li et al., 2025, Cheng et al., 2022), holding significant importance for human life and production activities. However, human activities within such formations—such as groundwater extraction and recharge, geothermal exploitation, CO₂ geological storage, and crude oil production (Kouaied et al., 2025, Liu et al., 2025)—typically involve fluid injection and withdrawal. Due to their low degree of cementation during formation, unconsolidated sandstones are highly susceptible to sand (and gravel) production under the action of formation fluid flow. This can lead to a series of problems: sand production alters formation permeability and exacerbates heterogeneity (Nie et al., 2014); it creates preferential flow paths, reducing sweep efficiency; sand particles clog wellbores and erode equipment (Amadi et al., 2024); and it compromises the long-term

stability of reservoirs or aquifers. Therefore, elucidating the mechanisms of sand production and the characteristics of residual fluid distribution under fluid-driven conditions within unconsolidated sandstone formations is of paramount scientific and engineering significance for predicting, preventing, and mitigating related engineering issues, thereby ensuring the safety and long-term stability of aquifers and reservoirs.

Significant research efforts in recent years have been focused on unconsolidated sandstone reservoirs, primarily encompassing sand production mechanisms, permeability evolution, clogging behaviors, and control strategies. The specific research directions and key findings from recent studies are summarized in Table 1. Despite these advancements, critical gaps remain in understanding the dynamic relationship between sand production and time, qualitative analysis of oil saturation in waterflooded zones, quantitative assessment of post-waterflood residual oil, and the impact of injector well placement on development efficiency in unconsolidated sandstone reservoirs.

Macro-scale experimental phenomena often require interpretation through micro-scale experiments, whose results, in turn, are frequently

^{*} Corresponding author.

E-mail address: zheyu.liu@cup.edu.cn (Z. Liu).

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Nomenclature	
R	red channel value
G	green channel value
B	blue channel value
Y	luminance value
U	chrominance blue value
V	chrominance red value
Cb	chrominance blue value
Cr	chrominance red value
IOM	image overlay method
S _{target}	sand grain migration area
NUM _{threshold}	number of blue pixels of a specific threshold
NUM _{all}	number of all blue pixels
S _{all}	whole image area

analyzed with the aid of image processing techniques. For example, Guanshi Liu et al. (Liu et al., 2025) employed image processing and machine learning methods for soil moisture content research. By developing a multi-Gaussian fitting-based color feature extraction approach combined with multi-source feature fusion and machine learning modeling, the prediction accuracy of soil moisture content was significantly improved. Golzar M. et al. (Golzar et al., 2018) investigated the transport of polyacrylic acid-stabilized iron oxide nanoparticles in porous media under saturated steady-state flow conditions by measuring tracer/nanoparticle concentrations within a two-dimensional model packed with glass beads using image processing techniques. S.Farshid Chini et al. (Chini and Amirfazli, 2011) proposed an image processing technique for measuring the contact angles of symmetric and asymmetric droplets by analyzing side-view images of droplets. This method eliminates the need for any liquid parameters, significantly reducing experimental errors. M. Mirzaei et al. (Mirzaei, 2017) utilized the Harris corner detector function to assign precise locations to contact points, setting goniometric masks at each given point and calculating heights, while applying a Gaussian-weighted averaging function to measure the contact angles of objects. LI Junjian et al. (Li et al., 2018) employed image processing techniques to separate particles and pore throats based on the principle of significant gray gradient differences between them. Fen Li et al. (Li et al., 2012) improved the accuracy of image segmentation by selecting the optimal gray value as the threshold segmentation point, using porosity and oil saturation as constraints. Arash Mehdizad et al. (Mehdizad et al., 2022) applied color extraction techniques to quantify the number of red pixels (dyed oil) before and after displacement, thereby analyzing the effects of clay swelling and particle migration on recovery efficiency. Syed Imran Ali et al. (Ali et al., 2023) tackled the problem of asphaltene deposition in oil fields by using image processing techniques to analyze the light spots produced by asphaltenes in crude oil. By leveraging the number of peaks to determine whether asphaltene precipitation was present, he successfully minimized errors caused by manual identification. Dong Xiaohu et al. (Dong et al., 2022). investigated the displacement efficiency in heavy oil reservoirs using visualized physical models. By applying digital image processing techniques, they extracted the swept region area and calculated the corresponding sweep efficiency. Grayscale-based image segmentation techniques are commonly applied to CT scans, enabling the separation of rock matrix from fluids. This approach facilitates the analysis of fluid displacement, saturation changes, and other dynamic processes during core flooding experiment (Sun et al., 2021, AlZahrani et al., 2025, Li et al., 2024, Zhao et al., 2025, Wang et al., 2025, He et al., 2024).

Image thresholding techniques are traditionally effective for images with limited and stable color distributions. However, extensive research has been conducted to address more complex scenarios involving richly colored images. Table 2 summarizes several common and relatively

Table 1
Current research directions and key findings in unconsolidated sandstone formations.

Studies	Research direction	Main conclusions
(Shakiba et al., 2020)	How to suppress sand production in unconsolidated sandstone.	Injecting low-salinity water with silica nanoparticles enhances oil recovery while significantly mitigating sand production.
(Zhou et al., 2025)	The mechanism of clogging near recharge wells during groundwater replenishment caused by seepage effects.	Clogging from seepage-induced particle migration away from wells can be addressed by reverse recharge to reverse the flow direction.
(Imqam et al., 2018)	Microgel migration in unconsolidated sandstones and its impact on flow resistance.	Gel particle retention increases injection pressure, suggesting that optimizing gel strength, concentration, and size can improve localized plugging and sweep efficiency.
(Wang et al., 2024)	Stress sensitivity of unconsolidated sandstone.	An increase in confining pressure exacerbates stress sensitivity damage. The primary site of stress sensitivity is within small and medium pores. Rocks with more developed pore throats and higher permeability exhibit stronger stress sensitivity and are more prone to damage.
(Xiong et al., 2018)	Compaction and sand production on fluid flow.	Early water injection helps maintain reservoir pressure and sustain oil production.
(Yang et al., 2024)	Pore blockage mechanisms in unconsolidated sandstones.	Sand migration and compaction jointly cause plugging in unconsolidated sandstone. Grain composition critically affects how particles migrate and transport.
(Xiong et al., 2018)	The impact of compaction on the productivity of unconsolidated sandstone reservoirs was analyzed.	Reveal irreversible reductions in porosity, permeability, and pore-throat size under compaction, with permeability declining most drastically.
(Shabdirova et al., 2022)	Sand production mechanisms.	The plastic zone comprises a high-permeability shear band and a low-permeability compaction zone; increasing porosity and permeability exacerbates sand production by facilitating particle migration between these zones.
(Perera et al., 2017)	Identify key sand production factors.	Sand production rates rise with increasing formation resistance but decrease with higher cementation.
(Zivar et al., 2019)	The impact of various parameters, on the final permeability after sand production.	Higher stress results in less sand production, leading to an increase in permeability caused by sanding.
(Chuanzhi et al., 2014)	Causes of large pore throats formed by water flooding in unconsolidated sandstone and their identification criteria.	A numerical reservoir model with time-dependent permeability and relative permeability curves was developed to determine large pore throat formation per grid cell, which is primarily caused by permeability increase due to water flooding erosion, based on established identification criteria.
(Ren et al., 2025)	CO ₂ storage in unconsolidated and coarse sandstones.	Over 80 % of CO ₂ can be sequestered in unconsolidated sandstones under optimal conditions, with storage efficiency twice that of coarse sandstones—a significant contribution to geological carbon storage.

Table 2
Current image processing techniques and their theory.

Studies	Methods	Principle
(Benaissa et al., 2024) (Kazmi et al., 2015)	The Express Green	Based on the RGB color space, this method capitalizes on the principle that the G (green) channel value is overwhelmingly dominant to enable the identification and extraction of green pixels.
(Xie et al., 2023) (Kang et al., 2024) (Win et al., 2025) (Ma and Yue, 2022)	Ostu's Method (The maximum inter-class variance method)	Automatically determines the optimal image threshold by maximizing inter-class variance to best separate pixels into two groups.
(Shaik et al., 2015) (Garcia-Lamont et al., 2018)	YUV and YCbCr model	The luminance and chrominance information of the image are separated. A target region is then identified and extracted by selecting an appropriate threshold.
(Jiang and Ye, 2024) (Aydın and Uğur, 2011)	Ant Colony Optimization (ACO)	The segmentation of the foreground from the background is achieved by configuring key parameters like the cluster number, ant population size, and iteration count (Li et al., 2023)

complex image processing methods along with their underlying principles.

While significant contributions have been made to the study of unconsolidated sandstone and the application of image processing techniques, considerable scope remains for developing novel image analysis methods and their application to unresolved research questions. This study first introduces a new image processing technique to extract and analyze data from images captured during physical experiments. The methodology is then applied to investigate four key aspects: (1) the relationship between sand production and time during water flooding; (2) the determination of a critical threshold based on the proposed method for confirming residual oil saturation; (3) the impact of sandstone heterogeneity on the distribution of residual oil saturation within the swept zone; and (4) the influence of reservoir heterogeneity on sweep efficiency and its implications for well placement strategies.

2. Materials and methods

2.1. Materials

During the experiment, transparent glass beads were used to simulate the rock framework of unconsolidated sandstone, with quartz sand serving as the interstitial material between the framework grains (Fig. 1a). The sizes of the glass beads and the mesh numbers of the quartz sand used in the experiment are provided in Table 3. The study involved displacement experiments on both homogeneous and heterogeneous unconsolidated sandstone models. The homogeneous model utilized glass beads of a single size, whereas the heterogeneous model incorporated glass beads of varied sizes—larger beads in high-permeability zones and smaller beads in low-permeability zones. As this research focuses on the impact of reservoir heterogeneity on the development of unconsolidated sandstone reservoirs, the permeability during the experiments was considered only in terms of relative differences between the models. After model preparation, crude oil obtained from the field was used to saturate the models; relevant crude oil parameters are listed in Table 3. The displacing phase consisted of formation water simulated with methylene blue dye, and related information on the formation water is provided in Table 3

During the experiment, the outlet was at atmospheric pressure, and a constant pressure differential of 0.05 MPa was maintained between the

Table 3
Physical properties of the fluids and solids employed in the experiments.

Simulated Formation Water	Salinity(mg/L)	Na ⁺ /K ⁺	11028.73
		Ca ²⁺	120.24
		Mg ²⁺	72.92
		Cl ⁻	14418.74
		SO ₄ ²⁻	1690.44
		HCO ₃ ⁻	3688.05
		CO ₃ ²⁻	0
		General Salinity	28654.95
		Viscosity (mPa.s)	1
		Density(g/cm ³)	0.873
Crude Oil	Viscosity (mPa.s)	85.9	
		Wax Content (%)	3.64
Glass Bead	Diameter(mm)	1,1.5,2,2.5,3	
		mesh	150-200

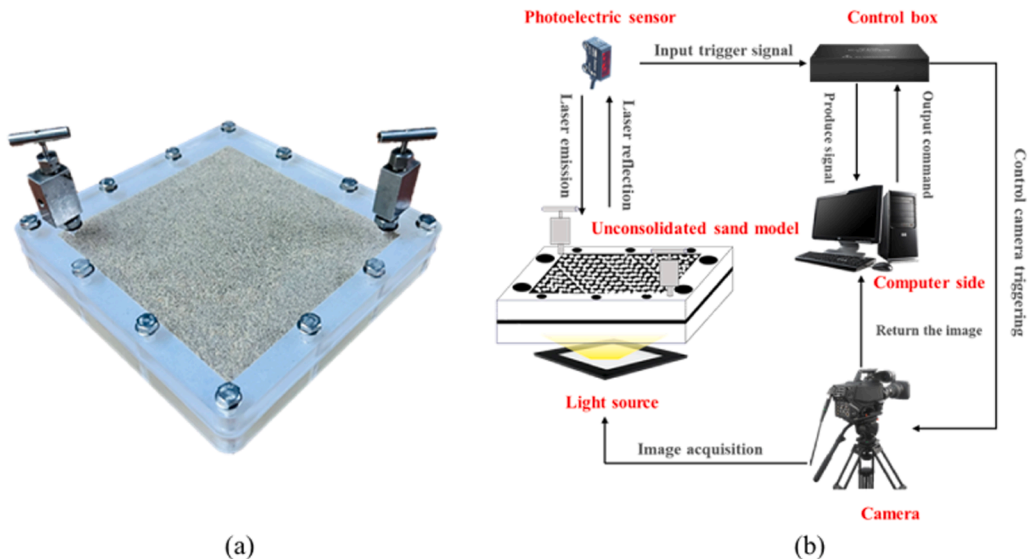


Fig. 1. (a) Micro-visualization model, (b) Image acquisition system.

inlet and outlet. Timed image acquisition was performed using a digital imaging system (Fig. 1b). Images were captured with a Sony camera at a resolution of 3480×2160 . The acquired images were subsequently processed using digital image analysis techniques to calculate relevant parameters.

2.2. Image overlay method (IOM)

This paper introduces a novel image processing method based on the RGB color space (Kolakoti and Chandramouli, 2025, de Carvalho Oliveira et al., 2022). By leveraging the characteristic that the blue channel (B) value in blue pixels is generally higher than both the red (R) and green (G) values, 254 blue threshold points ranging from 1 to 254 are defined. Taking a threshold of 50 as an example, a pixel is identified as blue if its B value is greater than or equal to 50, while both its R and G values are less than 50. The corresponding position in the logical matrix is then set to 1; otherwise, it is set to 0. This process generates a blue pixel extraction mask under the specified threshold condition, ultimately producing an image extracted under that criterion. With 254 blue thresholds applied, 254 distinct images can be extracted. Since the extraction results vary across different threshold levels, the 254 logical matrices are combined using a logical OR operation into a single unified logical matrix for image extraction. This enables effective identification of regions containing methylene blue in the image. The overall workflow is illustrated in Fig. 2.

$$B \geq \text{bluethreshold} \quad (1)$$

$$R < \text{bluethreshold} \quad (2)$$

$$G < \text{bluethreshold} \quad (3)$$

$$S_{\text{target}} = \frac{\sum_{i=1}^{255} \text{NUM}_{\text{threshold}}}{\text{NUM}_{\text{all}}} \times S_{\text{all}} \quad (4)$$

During the displacement process, the swept area includes zones with a high degree of scouring, oil-water mixed regions, and pure oil regions (Fig. 3).

In areas with higher scouring, blue brightness is elevated under bottom-light illumination, corresponding to higher grayscale values. In oil-water mixed regions, the presence of crude oil reduces blue brightness, resulting in lower grayscale values. As a result, the grayscale distribution of blue intensity across the swept area exhibits a wide range. According to experimental requirements, sample points within target regions were collected from the images, and the range of blue grayscale values in these samples was statistically determined. To further

minimize experimental error, the selected blue sample points were converted to grayscale. The K-means clustering (Bajpai et al., 2025, Hnin et al., 2024, Abdullahi et al., 2024) method was then employed to remove points with anomalous grayscale values by selecting an appropriate number of clusters. This process helped define the specific grayscale range corresponding to the blue pixels in the target area, thereby enhancing the accuracy of the experimental results. The workflow of the K-means method is as follows.

- (1) Determine the number of clusters and the number of cluster centers.
- (2) Sequentially examine each sample and assign it to the nearest cluster based on Euclidean distance.
- (3) Calculate the center of each cluster.
- (4) Repeat the iteration until convergence is achieved.

The initially extracted images were converted to grayscale, and different target regions were segmented based on the grayscale ranges of the sample points. It is worth noting that, in highly scoured regions, the reflective properties of glass beads cause their surroundings to appear with relatively high blue intensity. Therefore, an accurate representation of the swept area should be obtained by subtracting the glass bead periphery from the extraction results described above. The overall procedure is illustrated in Fig. 4.

3. Results and discussion

This chapter begins by validating the effectiveness of the proposed method based on the amount of sand produced during the experiments, while also demonstrating its advantages compared to existing approaches. Subsequently, it analyzes the relationship between sand production and both time and heterogeneity during the water flooding process. The analysis is further extended to examine the correlation between residual oil saturation and heterogeneity within the swept zone. Based on the observed patterns of residual oil saturation change, a threshold for determining residual oil saturation is established. Finally, the relationship between sweep efficiency and heterogeneity is discussed.

3.1. Effectiveness and novelty of image overlay method

Due to the weak cementation of loose sandstone, sand grains are readily dislodged and transported to the outlet during water flooding. The cumulative sand production throughout the flooding process was measured experimentally. Subsequently, the proposed image-based

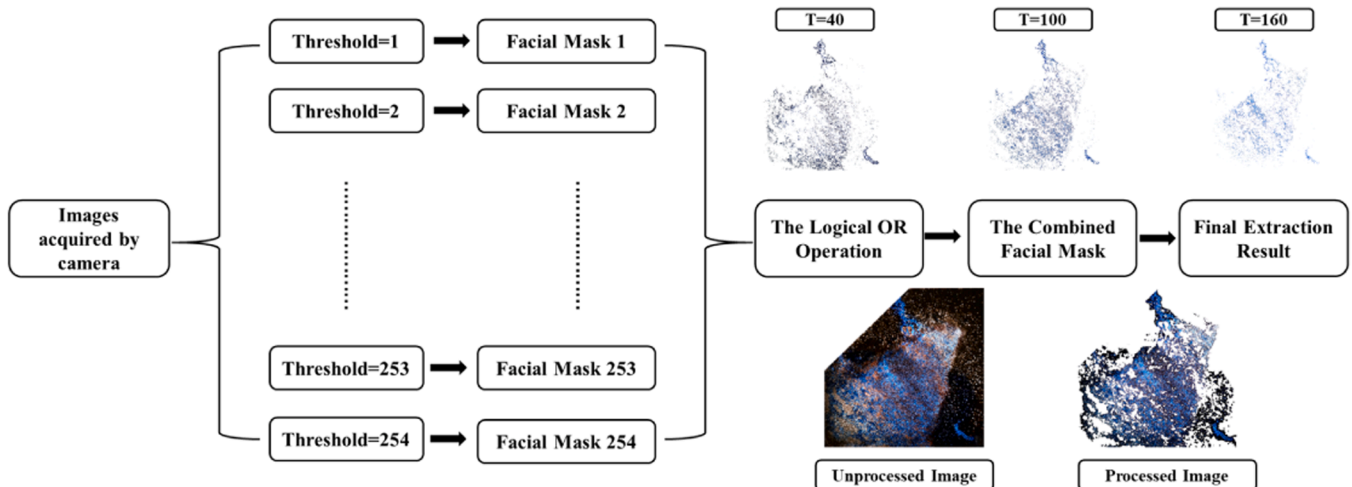


Fig. 2. Schematic diagram of the preliminary extraction of the methylene blue solution flooding area.

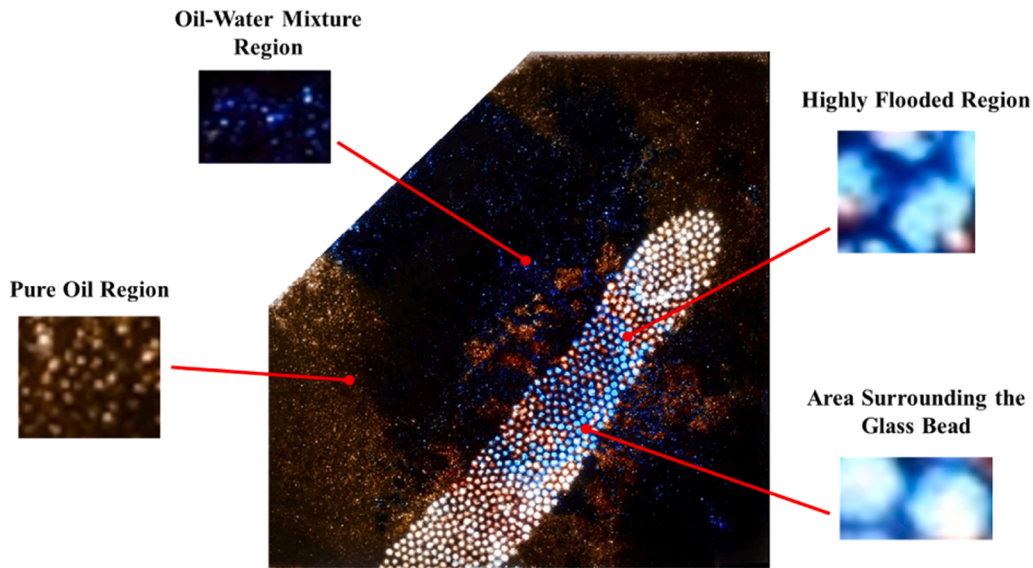


Fig. 3. Different types of target regions.

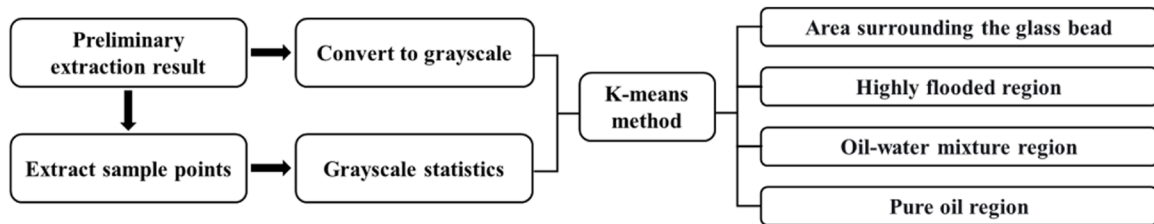


Fig. 4. Schematic diagram of result refinement processing.

analysis method was applied to quantify sand production from the final image. The sand production quantity, obtained by the image overlay method, was calculated from the area of sand grain migration extracted by the method and the diameter of the glass beads used in the experiment. A comparison between the two measurement results (Fig. 5) indicates that the sand production volume estimated through image processing is generally slightly lower than the physically measured value. This discrepancy arises because the image method primarily

accounts for sand loss in areas exhibiting a high degree of scouring, while sand production also occurs within oil-water mixed regions, leading to an underestimation. Nevertheless, the overall relative error remains below 0.63 %, demonstrating that the image overlay method proposed in this study remains effective and meaningful.

Through the application of various methods to process the images in this experiment, the following conclusions can be drawn: The ExG (Excess Green) algorithm operates on the principle that green pixels exhibit a dominant G-channel value in the RGB color space, enabling the identification of green regions. A similar approach can be applied for blue region detection. However, this method demonstrates significant limitations when dealing with images containing rich color variations (Fig. 6).

Otsu's method separates foreground from background by selecting an optimal threshold that maximizes inter-class variance. While it performs satisfactorily when the target regions are concentrated, it often fails to achieve high segmentation accuracy (Fig. 7b). The YUV and YCbCr color models decouple luminance information from chrominance components. Color segmentation is subsequently performed by applying appropriate thresholds. Similar to Otsu's method, the performance of these models is contingent upon threshold selection, inevitably introducing certain degrees of error (Fig. 7c and Fig. 7d). The Ant Colony Algorithm, which involves the configuration of multiple parameters and substantial computational overhead, is also more suitable for scenarios with clustered target areas. Its effectiveness markedly decreases when target regions are dispersed (Ariza-Sentís et al., 2024, Calle et al., 2017) (Fig. 7e).

The proposed method effectively addresses the aforementioned limitations. It eliminates the need for manual threshold selection by leveraging the dominant B-channel value of blue pixels in the RGB color space. By exhaustively evaluating all 254 potential thresholds, the

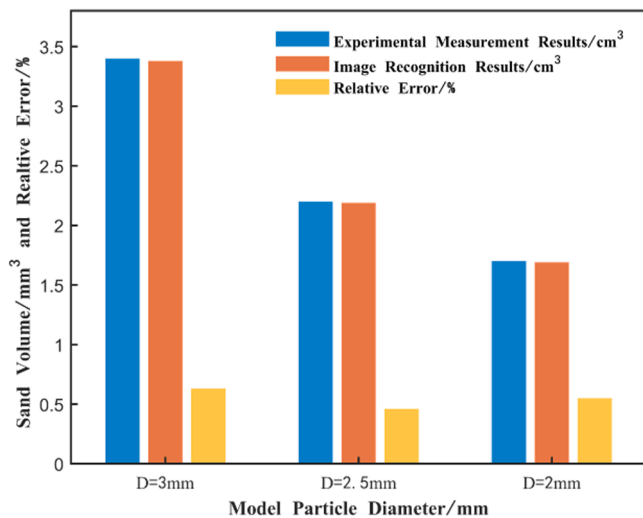


Fig. 5. Comparison between measured and image-based sand production during water flooding in different models.

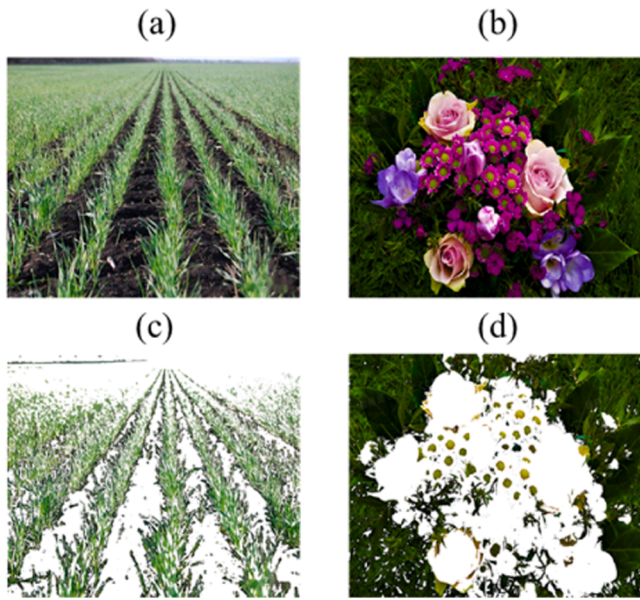


Fig. 6. (a, b) Original images, (c, d) Results of the excessive green algorithm.

method mitigates human-induced biases associated with subjective thresholding. Furthermore, it remains robust even when target regions are widely distributed. The algorithm is computationally efficient, requiring minimal runtime, thereby offering a practical and scalable solution for color-based image segmentation.

3.2. Contrastive characteristics of sand transport area between homogeneous and heterogeneous sandstone

As shown in Fig. 8, in heterogeneous unconsolidated sandstone

systems, the sand particle migration area ranges from 37.8 to 90.3 cm² (Fig. 8a), while in homogeneous unconsolidated sandstones, it ranges from 22.7 to 41.4 cm² (Fig. 8b). This phenomenon demonstrates that reservoir heterogeneity significantly influences the propagation range of waterflood fronts, thereby regulating sand particle migration behavior. Experimental data indicate that the sweep efficiency of heterogeneous unconsolidated sandstones is generally higher than that of homogeneous systems. The underlying mechanism can be attributed to the following: although both types of core samples exhibit similarly high average porosity (Xiao et al., 2024) the spatial heterogeneity of pore structures in heterogeneous reservoirs induces dynamic diversion of waterflood fronts, markedly increasing the branching probability of displacing fluids within porous media (Lee et al., 2022). Regression analysis reveals that the sand transport area in heterogeneous unconsolidated sandstone follows a quadratic polynomial relationship with time ($R^2 > 0.95$), which provides critical insights for sand production prediction in such reservoirs. Quantitative analysis reveals that the rate of change in sand particle migration area for heterogeneous cores (0-4) (Fig. 8c) is significantly greater than that for homogeneous cores (0-2) (Fig. 8d). This order-of-magnitude difference in variation rates reflects two key dynamic behaviors during the displacement process: abrupt diversion of waterflood fronts and proliferation of dominant flow channels. The key finding of this study is that moderate reservoir heterogeneity can effectively improve development performance in unconsolidated sandstone reservoirs by promoting fluid branching effects at the microscale. This understanding provides a new theoretical basis for optimizing development strategies in high-permeability unconsolidated sandstone reservoirs.

3.3. Comparative study on remaining oil distribution patterns between homogeneous and heterogeneous unconsolidated sandstone post waterflooding

The grayscale characteristics of the blue color near glass beads in water-flooded zones were extracted by k-means method, the liquid

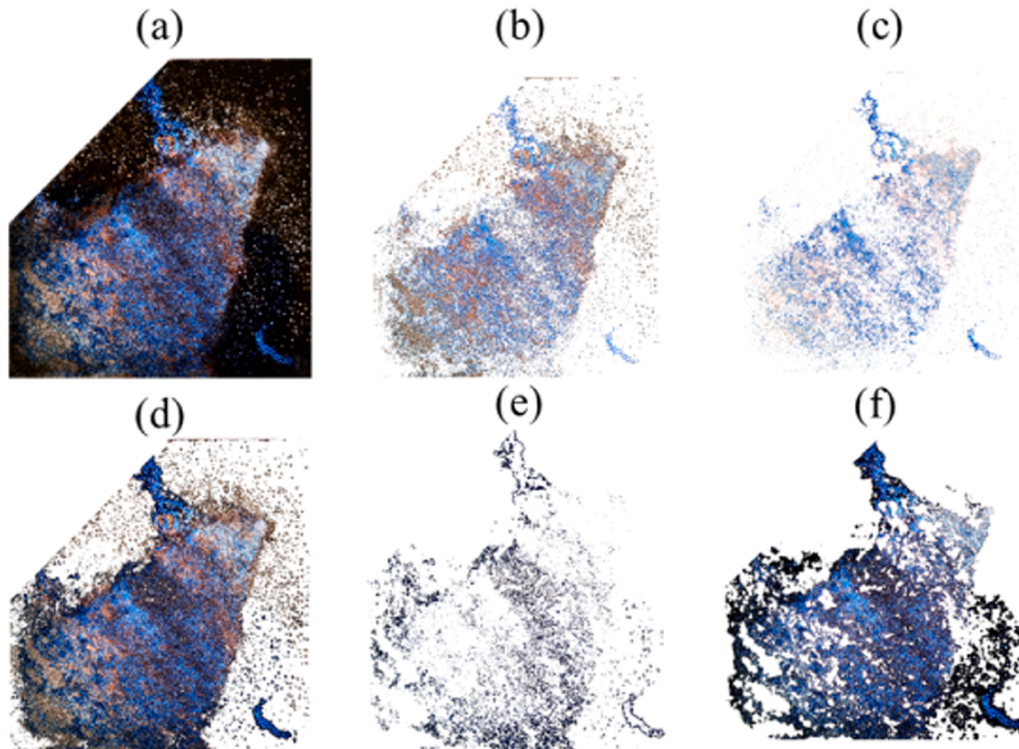


Fig. 7. (a) Original image, (b) Result extracted by the Otsu method, (c) Result extracted using the YUV model, (d) Result extracted using the YCbCr model, (e) Result extracted by the ant colony algorithm, (f) Result extracted by the image superposition method.

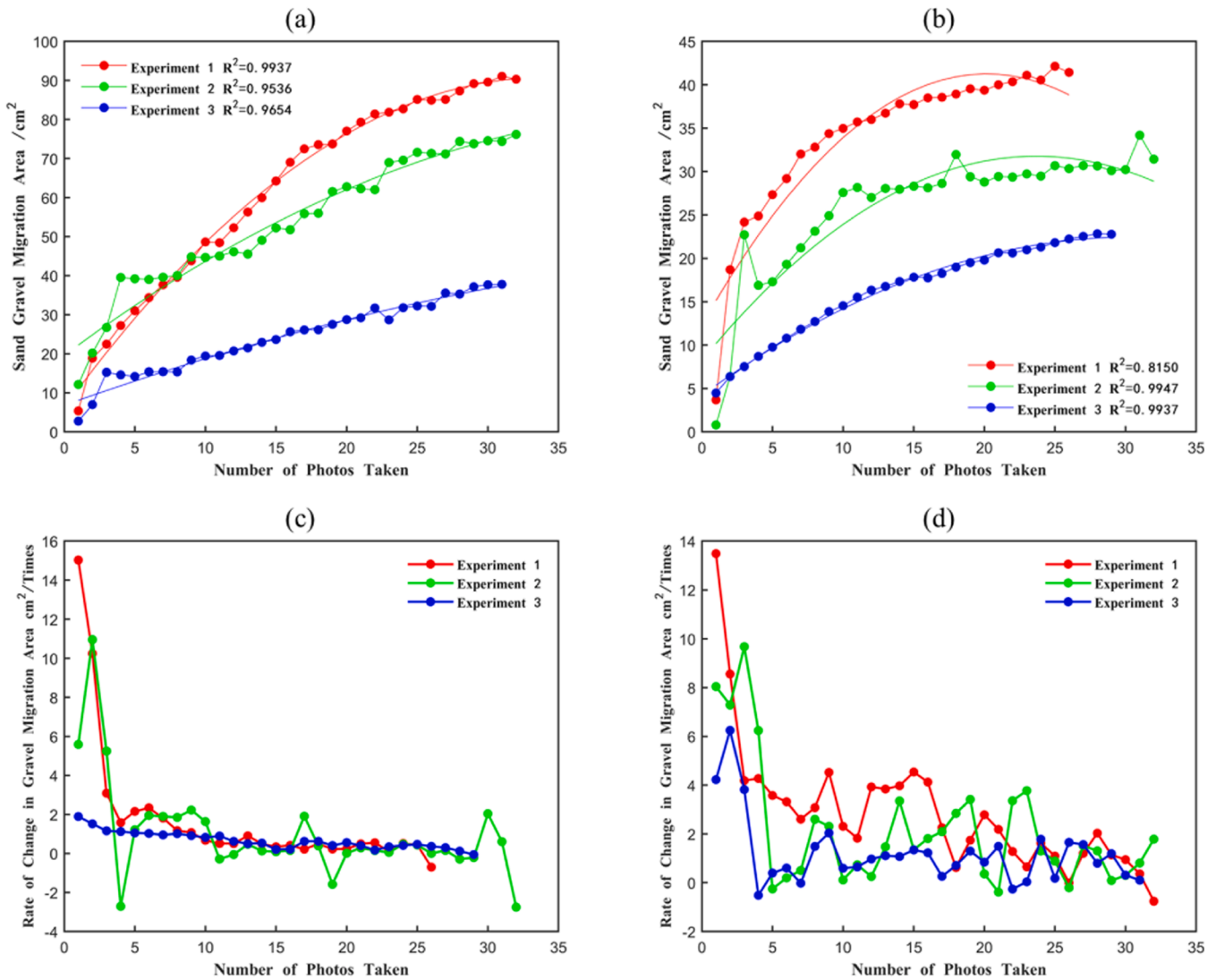


Fig. 8. (a) Relationship between gravel migration area and time in heterogeneous sandstone, (b) Relationship between gravel migration area and time in homogeneous sandstone, (c) Relationship between the rate of change in gravel migration area and time in heterogeneous sandstone, (d) Relationship between the rate of change in gravel migration area and time in homogeneous sandstone.

distribution within the swept area was obtained by removing the aforementioned blue pixels. Due to interference from crude oil and interstitial sand grains, additional colors were introduced into the methylene blue solution, resulting in varying brightness distributions across the overall color. In regions with higher residual oil content, the interference with the methylene blue color was more significant, causing the overall blue hue to appear darker. In contrast, in areas with stronger scouring effects where residual oil content was lower, the brightness of the blue color was less affected, making the overall blue hue appear brighter. As shown in Fig. 9, it displays the range of blue brightness variations in water-flooded zones of heterogeneous unconsolidated sandstone. The relatively uniform distribution of blue pixels across different brightness levels indicates the coexistence of both high and low residual oil saturation regions after water flooding in heterogeneous sandstone. Fig. 9c and Fig. 9d present the blue brightness distribution in water-flooded zones of homogeneous unconsolidated sandstone, where the blue brightness is mainly concentrated in the 0-50 range. This suggests that for homogeneous sandstone, the variation in residual oil saturation within the swept zone is small after water flooding (Du et al., 2024). Theoretically, for homogeneous unconsolidated sandstone, the relatively uniform physical properties lead to consistent oil displacement by injected fluids during water flooding, resulting in a

concentrated distribution pattern of residual oil saturation within the swept zones. In contrast, heterogeneous unconsolidated sandstone exhibits marked non-uniform flushing characteristics during water flooding due to significant variations in pore-throat connectivity. Image analysis reveals that grayscale variations directly reflect these flushing disparities: higher grayscale values correspond to more effective displacement, while lower values indicate suboptimal flushing efficiency. Consequently, enhancing the uniformity of flushing efficiency within swept zones of heterogeneous unconsolidated sandstone remains a critical scientific challenge in current research aimed at improving oil recovery efficiency.

3.4. Threshold image analysis for quantifying post-waterflood bulk residual oil saturation in unconsolidated sandstone

Within waterflood-swept zones, regions subjected to more intensive flushing exhibit lower residual oil saturation (S_{or}), whereas less effectively swept areas retain higher S_{or} . The residual oil predominantly exists in various morphological forms, including filamentary, corner-bound, and cluster-type distributions etc. Accurately determining the average residual oil saturation across the entire swept zone remains a critical challenge in optimizing oil recovery efficiency. During

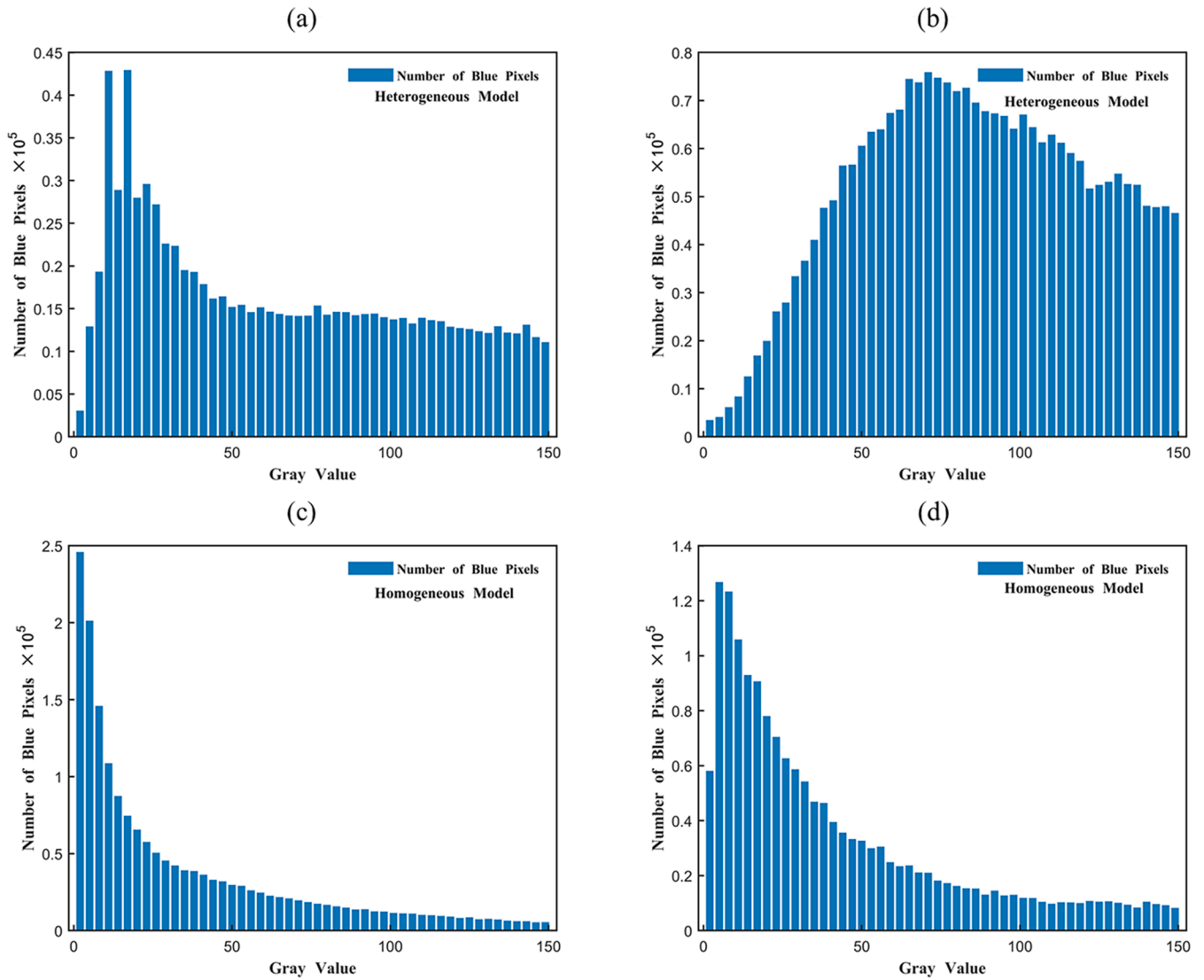


Fig. 9. (a, b) Blue pixel distribution in flooded heterogeneous sandstone, (c, d) Blue pixel distribution in flooded homogeneous sandstone.

waterflooding processes, the expansion of water-invaded areas leads to a progressive increase in residual oil saturation (S_{or}) within swept zones, while the oil saturation in unswept regions exhibits a corresponding decline (Fig. 10). As the displacement process continues, the overall oil saturation of the reservoir demonstrates a consistent decreasing trend. Consequently, the combined oil saturation of swept and unswept zones must follow a monotonically decreasing relationship. The fundamental characteristic enables the application of grayscale threshold analysis to precisely determine the critical threshold value distinguishing water-dominated zones from oil-water mixed regions within the swept area. The established methodology provides an accurate approach for residual oil saturation quantification, offering significant implications for enhanced oil recovery (EOR) strategies.

As illustrated in Fig. 11, the selection of grayscale threshold values significantly influences saturation curve characteristics. When an excessively high threshold is applied, the oil saturation curve demonstrates an increasing trend. This phenomenon occurs because the elevated threshold misclassifies portions of the water-swept zones as oil-bearing regions, resulting in an overestimated oil saturation increase that surpasses the actual decrease in unswept zones. Conversely, an optimally selected threshold produces a quasi-parabolic oil saturation profile, where distinct grayscale contrast (bright-dark differentiation) during specific displacement stages induces a reversal in saturation

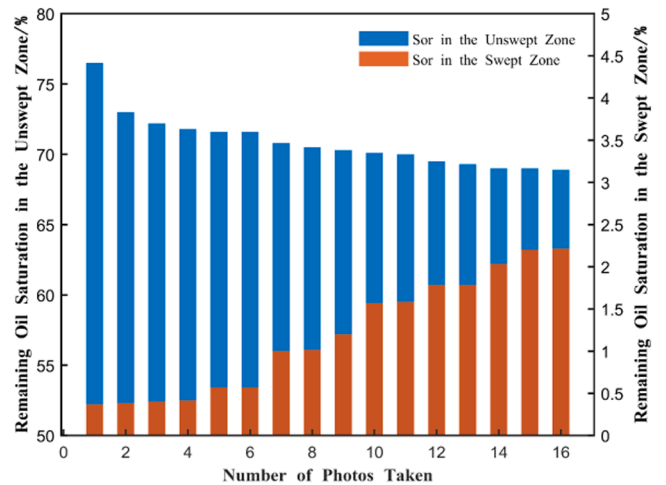


Fig. 10. Variation relationship of oil saturation between swept and unswept zones.

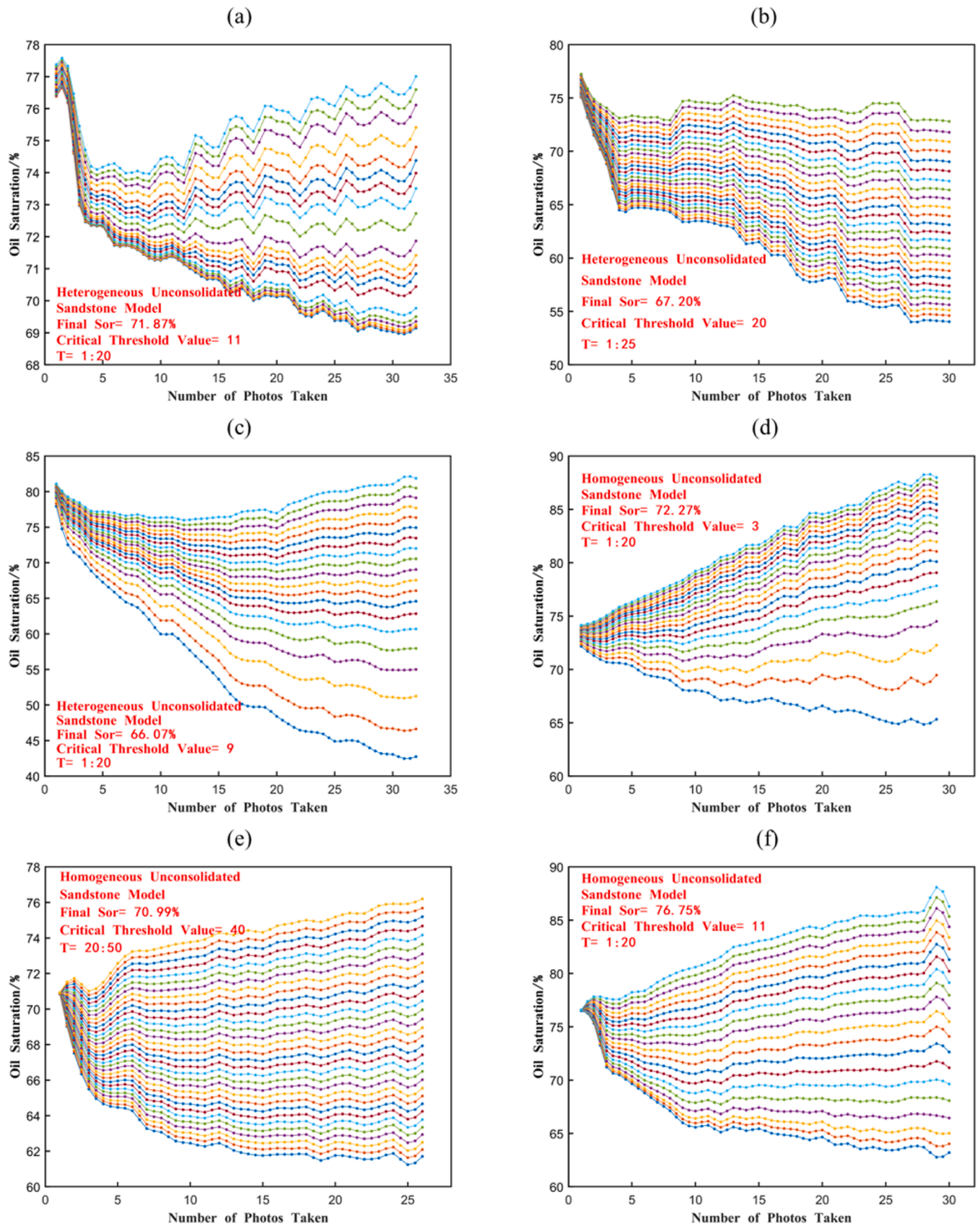


Fig. 11. (a, b, c) Critical threshold identification results for the oil-water mixing zone in the heterogeneous model, (d, e, f) Critical threshold identification results for the oil-water mixing zone in the homogeneous model.

change rates between zones. An undersized threshold presents different challenges, failing to adequately identify residual oil in swept zones and consequently reducing the measured saturation increment. While this scenario yields a decreasing trend curve, the substantial computational errors introduced by threshold underestimation render the results unreliable. Therefore, the true critical threshold should correspond to the inflection point where the saturation profile transitions from parabolic to monotonically decreasing. Experimental data (Fig. 11) reveal consistently low primary recovery factors for both homogeneous and heterogeneous unconsolidated sandstone formations. Quantitative analysis shows final residual oil saturations ranging: 66.07–71.87 % for heterogeneous models and 70.99–76.65 % for homogeneous counterparts. These results demonstrate that reservoir heterogeneity contributes measurably to enhanced oil recovery, with heterogeneous systems exhibiting marginally better recovery performance. The findings provide valuable insights for threshold-based saturation analysis and highlight the influence of geological heterogeneity on displacement efficiency.

3.5. Impact of heterogeneity on crude oil recovery efficiency

As illustrated in Fig. 12, under constant displacement pressure conditions, the water flooding sweep efficiency in heterogeneous unconsolidated sandstones is significantly greater than that in homogeneous systems. Experimental measurements show that the maximum sweep area in homogeneous sandstone is 61.7403 cm² (Fig. 12e), whereas even the minimum sweep area in heterogeneous sandstone reaches 65.1502 cm²

cm² (Fig. 12b). This demonstrates that reservoir heterogeneity enhances fluid flow dynamics by promoting more efficient displacement. From the perspective of flow channel development, homogeneous sandstones predominantly develop three dominant flow paths, whereas in heterogeneous systems, permeability contrasts induce fluid redistribution, leading to the formation of multiple branching preferential channels, thereby improving sweep efficiency. Notably, when the injection point is located in high-permeability zones, the sweep areas increase substantially to 92.0678 cm² (Fig. 12a) and 104.0768 cm² (Fig. 12c), significantly outperforming the case where injection occurs in low-permeability zones (65.1502 cm², Fig. 12b). The results indicate that placing injectors in high-permeability regions can substantially enhance oil recovery. Furthermore, a positive correlation exists between the extent of low-permeability zone contact and the sweep area during displacement. These findings provide critical theoretical support for the development of unconsolidated sandstone reservoirs, suggesting that injection well placement should be optimized based on reservoir heterogeneity characteristics to maximize flooding efficiency.

4. Conclusion

This study investigates gravel transport mechanisms and residual fluid distribution in hydrogeological systems by injecting formation water into artificially constructed unconsolidated sandstone models. A threshold-insensitive image analysis technique is developed to quantitatively characterize gravel migration pathways and residual oil saturation. Comparative experiments with homogeneous models elucidate

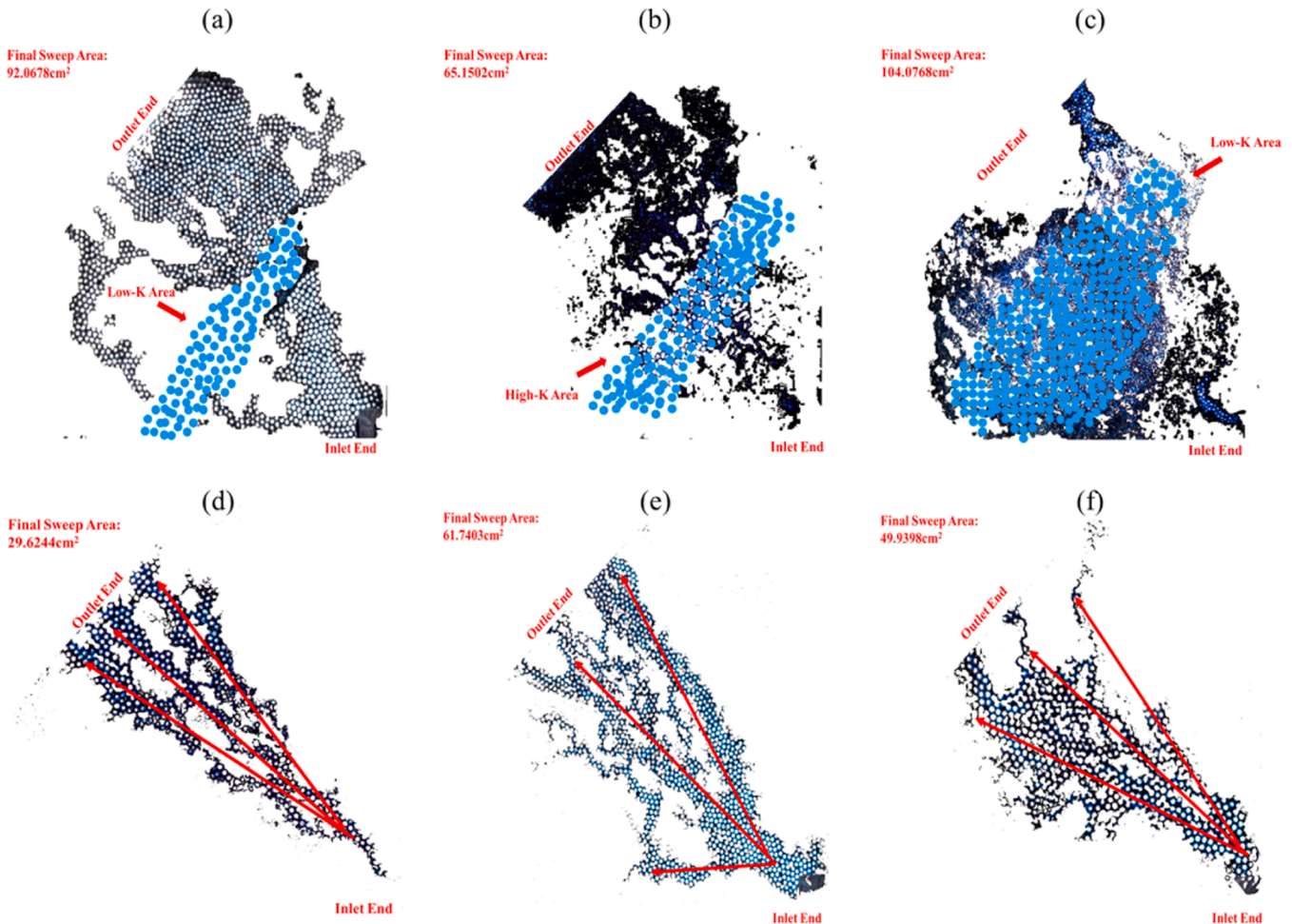


Fig. 12. (a, b, c) Sweep area of water flooding in heterogeneous unconsolidated sandstones, (d, e, f) sweep area of water flooding in homogeneous unconsolidated sandstones.

the influence of heterogeneity on fluid flow behavior.

The results indicate pronounced differences in production characteristics between heterogeneous and homogeneous sandstones. Heterogeneous systems exhibit substantially larger sand production areas (37.8 cm² to 90.3 cm²) and sweep areas (65.15 cm² to 104.07 cm²) compared to homogeneous systems (22.7 cm² to 41.4 cm² and 29.62 cm² to 61.74 cm², respectively). This divergence is attributed to the frequent diversion of the flood front caused by heterogeneity, which enhances both sand mobilization and volumetric sweep efficiency.

Furthermore, residual oil saturation within swept zones displays distinct distribution patterns: a narrow range in homogeneous sandstones (70.99 % to 76.65 %) suggests consistent displacement, whereas a broader range in heterogeneous sandstones (66.07 % to 71.87 %) reflects uneven fluid advancement. The observed saturation variation provides a criterion for determining optimal segmentation thresholds in image analysis. Importantly, strategic placement of injectors in high-permeability zones of heterogeneous formations is shown to significantly improve sweep efficiency.

This work advances the management of fluid injection processes in aquifer and reservoir systems by: enhancing understanding of coupled particle-fluid transport; establishing a theoretical basis for predicting permeability evolution, preferential flow path development, and residual phase distribution; introducing an innovative imaging method for pore-scale process quantification; and providing practical guidelines for injector placement in reservoir recharge, environmental remediation, and subsurface energy storage applications.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CRedit authorship contribution statement

Maozhu Li: Writing – original draft, Methodology, Formal analysis, Data curation. **Yiqiang Li:** Writing – review & editing, Funding acquisition. **Zhipeng Wang:** Methodology, Funding acquisition. **Xinji Du:** Resources. **Lei Wang:** Resources. **Jinxin Cao:** Methodology. **Zheyu Liu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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